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Project Title	Product and Category Management System
Week	Week 2

WEEK-2 TASK

Task Name

Product and Category Management System

Abstract:

The Product and Category Management System is a database-based application designed to manage products and their categories efficiently. It helps businesses organize product information, track categories, and maintain accurate records. The system uses SQL for storing, retrieving, updating, and deleting product data. By maintaining a structured relationship between products and categories, the system improves inventory organization, reduces data redundancy, and enhances decision-making.

Introduction:

In modern businesses, managing products and categories is an important task. A Product and Category Management System provides a centralized platform to store product details such as product name, price, quantity, and category information. The system enables users to categorize products properly, making it easier to search, update, and manage inventory records. SQL databases are used to ensure data consistency, integrity, and efficient data retrieval.

Objectives:

- To store and manage product details efficiently.
- To organize products into different categories.
- To perform CRUD (Create, Read, Update, Delete) operations on products and categories.
- To reduce data redundancy through database normalization.
- To provide quick access to product information.
- To improve inventory management and reporting.

SQL Code:

Create Database

```
CREATE DATABASE ProductManagement;
```

```
USE ProductManagement;
```

Create Category Table

```
CREATE TABLE Category (  
    CategoryID INT PRIMARY KEY,  
    CategoryName VARCHAR(50) NOT NULL  
);
```

Create Product Table

```
CREATE TABLE Product (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(100) NOT NULL,  
    Price DECIMAL(10,2),  
    Quantity INT,  
    CategoryID INT,  
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
);
```

Insert Data into Category Table

```
INSERT INTO Category VALUES  
(1, 'Electronics'),  
(2, 'Clothing'),  
(3, 'Books');
```

Insert Data into Product Table

```
INSERT INTO Product VALUES  
(101, 'Laptop', 55000.00, 10, 1),  
(102, 'Mobile Phone', 25000.00, 20, 1),  
(103, 'T-Shirt', 500.00, 50, 2),
```

(104, 'Python Programming Book', 700.00, 30, 3);

Display Products with Categories

```
SELECT P.ProductID,  
       P.ProductName,  
       P.Price,  
       P.Quantity,  
       C.CategoryName  
FROM Product P  
JOIN Category C  
ON P.CategoryID = C.CategoryID;
```

Update Product Price

```
UPDATE Product  
  
SET Price = 60000.00  
  
WHERE ProductID = 101;
```

Delete a Product

```
DELETE FROM Product  
  
WHERE ProductID = 104;
```

Result:

The Product and Category Management System was successfully implemented using SQL. The system efficiently stored product and category information, maintained relationships between tables using foreign keys, and supported operations such as insertion, updating, deletion, and retrieval of records. The output displayed products along with their corresponding categories accurately.

Sample Output

ProductID	ProductName	Price	Quantity	CategoryName	
101	Laptop	55000	10	Electronics	
102	Mobile Phone	25000	20	Electronics	
103	T-Shirt	500	50	Clothing	

104	Python Programming Book	700	30	Books	
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Conclusion:

The Product and Category Management System provides an effective solution for managing product inventories and category information. By using SQL databases, the system ensures efficient data storage, integrity, and quick retrieval of information. This project demonstrates the practical application of database management concepts and can be extended with additional features such as user authentication, stock monitoring, and sales management for real-world business applications.